

Adaptive Traffic Signal Cycle Control for Green City Traffic Management

Akshita Gupta¹, Anant Agarwal¹, and Vivek Ashok Bohara¹

¹Wirocomm Research Group, Department of Electronics & Communication Engineering,
Indraprastha Institute of Information Technology Delhi (IIITD), New Delhi 110020, India.

Email: akshitag@iiitd.ac.in, ananta@iiitd.ac.in, vivek.b@iiitd.ac.in

Abstract—Traffic congestion at road intersections is one of the critical challenges for vehicular traffic systems. In the urban environment, this traffic congestion results in an increase in the travel time, more driver fatigue, higher pollution, etc. Consequently, traffic congestion not only exacerbates travel delays but also impacts the environment and human health. As a consequence, the implementation of adaptive traffic signal cycle control to alleviate traffic congestion is of paramount importance. In this work, we use Simulation of Urban MObility (SUMO) based traffic simulation to control traffic signal cycle for smart cities. We propose a deep reinforcement learning (DRL)-based adaptive traffic signal cycle control that aims to minimize carbon dioxide (CO₂) emissions while simultaneously decreasing the waiting time of the drivers. The results showcase that with the use of a DRL-based adaptive traffic signal control cycle, specifically, the average waiting time of the vehicles increases by 15.58% as the number of vehicles increases from 1750 to 2000. Furthermore, the average CO₂ emissions of the vehicles increase by 25% as the number of vehicles increases from 1750 to 2000.

Index Terms—Traffic signal cycle, Traffic management, Greenhouse effect, Driver's waiting time, Deep Reinforcement Learning

I. INTRODUCTION

For a long time, traffic congestion has been recognized as a social and economic hindrance that has negatively impacted the quality of life, fuel consumption, air quality, and human productivity. These factors further contribute to a variety of environmental and social issues, including a higher rate of vehicular accidents, a reduced rate of productivity, energy waste, and poor air quality. The monitoring of traffic congestion is contingent upon the effective and efficient operation of traffic signals and their regulation. The primary challenge that the current intelligent transportation field is confronted with is the development of effective and precise rules for complex and vast traffic data. A smart traffic signal employs the most recent technology to enhance traffic flow by incorporating a data collection and processing network. This network processes and transmits the information to an existing network, which subsequently adjusts to the changing traffic parameters and provides the traffic controller with real-time signal timings. In the research field, corresponding control means are provided to make effective control decisions for the field of traffic manage-

ment, thereby preventing the continuous aggravation of traffic congestion and quickly dissipating the congested sections when combined with the characteristics of the current road structure.

The processing of actual traffic data is a challenging task for general methods and this leads to the utilization of learning-based mechanisms for managing the traffic signal cycle. Some of the adaptive traffic signal cycle control strategies proposed in the literature include [1]-[2]. Specifically, the authors in [1] discussed the adaptive traffic control system that utilized real-time network monitoring to manage and control the traffic flow. Further, the authors also discussed the current situation of severe traffic congestion in cities and multiple technological enhancements, including real-time data collection and data processing techniques that will result in a reduction in commuter travel time, and the number of hours spent in traffic, as well as a better travel experience for commuters. The authors in [3] considered pedestrian flow traffic along with vehicular traffic. The authors utilized a stochastic hybrid system model to propose an optimal sharing policy to achieve fairness in average waiting periods. This policy is based on the partially observed contents of pedestrian and vehicle delays and several threshold parameters that are applied to the light cycles. Further, the authors in [4] proposed a region adaptive traffic signal control strategy that utilized a double dueling deep Q network (3DQN). The authors proposed two strategies, namely, trust region state and standardized reward, to improve the performance of intersection control in an urban environment. [5] proposed a method to predict the traffic condition and control traffic congestion. The authors utilized improved particle swarm optimization and long short-term memory (LSTM) networks to predict traffic conditions. Depending on the predicted traffic conditions, a traffic congestion control algorithm is proposed. In [2], the authors proposed an optimal signal changing time that is broadcasted one cycle in advance, allowing automated vehicles to adjust their speed to cross the intersection with the least amount of stop-delay. The proposed strategy results in an increase in the average speed of the vehicles and reduces the traffic density and the number of idle vehicles. Further, the authors also analyzed the efficacy of the proposed strategy

on fuel consumption and the rate of carbon dioxide (CO₂) emissions in the vicinity of the intersection.

As evident from the above works, the existing works have not utilized the CO₂ factor along with waiting time to adaptively control traffic signal cycle for green city traffic management. This paper utilized a deep reinforcement learning (DRL) algorithm to control the traffic signal cycle at a single four-way intersection. The proposed algorithm utilizes the simulations that are agent-based. Specifically, we utilize Simulation of Urban MObility (SUMO) [6] simulator, which provides a synthetic yet realistic environment for exploring the outcomes of potential regulatory actions. Consequently, we were able to establish a plausible set of observable environmental characteristics, regulate the traffic lights in accordance with the learning agent's decisions, and utilize certain statistics collected by SUMO to describe the overall traffic flow and, as a result, establish the reward for the actions taken by the traffic light control agent.

The rest of this article is organized as follows. In Section II, we introduce the system model and provide a discussion on traffic light management with RL approaches. We present numerical results in Section III. Finally, Section IV concludes this article.

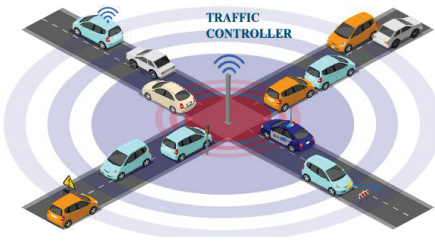


Fig. 1: System Model for traffic cycle control

II. SYSTEM MODEL

The system model for the traffic control cycle for green city traffic management is shown in Fig. 1. We consider a single four-way intersection regulated by traffic lights. Each of the four lanes has both incoming and outgoing traffic. The traffic controller takes into account the driver's waiting time and CO₂ emissions to control the traffic signal cycle.

A. Reinforcement Learning

In a reinforcement learning (RL) problem, an autonomous agent observes the environment and perceives a state s_t , which is the state of the environment at time t . Subsequently, the agent selects an action that results in the environment transitioning to the state s_{t+1} . The agent receives a reward r_{t+1} following the environment transition, which indicates the agent's performance in relation to a performance measure. The agent's objective is to learn the policy π^* that maximizes the cumulative expected reward obtained as a result of actions taken while

adhering to π^* . The typical reinforcement learning cycle is as illustrated in Figure 2.

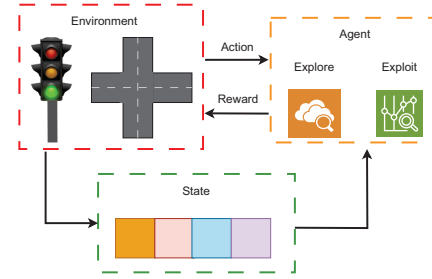


Fig. 2: RL framework for traffic light control

In the reinforcement learning framework in the traffic signal control cycle, one or more autonomous agents are tasked with maximizing the efficacy of traffic flow that passes through one or more intersections controlled by traffic lights. Several factors motivate the utilization of RL for traffic signal control, namely, RL agents can adapt to various situations (e.g., road accidents and inclement weather). RL agents can self-learn without supervision or prior knowledge of the environment, and the agent only requires a simplified model of the environment (mainly related to the state representation), as the agent learns using the system performance metric (i.e., the reward) [7].

The RL framework for the traffic signal cycle control is shown below:

1) *State Representation*: The agent's state is typically represented by the symbol s_t and provides a representation of the environment's state during a specific agent step t . The state must provide the agent with adequate information regarding the distribution of vehicles on each road in order to enable them to learn how to optimize traffic operations. We utilized discretized cells along the length of each lane running through the intersection. The cells in the lane are assigned to cells in a vector, which indicates 1 if a vehicle is present in the lane cell and 0 otherwise.

2) *Action State*: The agent's potential actions are identified by the action set. The agent is the traffic signal system, and performing an action activates a green phase for a specific number of lanes for a predetermined duration, selecting from a predetermined list of green phases. The yellow time is set at 4 seconds, while the green time is set at 10 seconds in this paper [7]. The action set encompasses all feasible actions that the agent may execute. These states are defined as follows:

- 1) North-South Advance (NSA): The green phase is active for vehicles that are in the north and south arm and wish to proceed straight or turn right.
- 2) North-South Left Advance (NSLA): The green phase is active for vehicles that are in the north and south arm and wish to turn left.

- 3) East-West Advance (EWA): The green phase is active for vehicles that are in the east and west arm and wish to proceed straight or turn right.
- 4) East-West Left Advance (EWLA): the green phase is active for vehicles that are in the east and west arm and wish to turn left.

Thus, the action state comprises of: $A = \{NSA, NSLA, EWA, EWLA\}$

3) *Reward Function*: The reward in reinforcement learning is the feedback from the environment that the agent receives after selecting an action. The agent employs the reward to comprehend the outcome of the action taken and enhance the model for future action decisions. Consequently, the compensation is an essential component of the learning process. The reward typically has two potential values: positive and negative. A positive reward is produced as a result of positive actions, while a negative reward is produced by negative actions. The objective of this application is to optimize the traffic movement through the intersection over time. To accomplish this objective, the reward must be derived from a performance measure of traffic efficiency, enabling the agent to determine whether the action taken has decreased or increased the efficiency of the intersection. This paper introduces two reward functions that employ two traffic measures that are marginally distinct, namely, the waiting time of the driver and CO₂ emissions.

$$RF = \sum_{v=1}^V [(T_{v,t-1} - T_{v,t}) + \alpha (CE_{v,t-1} - CE_{v,t})], \quad (1)$$

where V is the total number of vehicles, $T_{v,t}$ is the waiting time of vehicle v driver at t^{th} time slot and $CE_{v,t}$ is the CO₂ emissions of vehicle v driver at t^{th} time slot.

4) *Deep Q-Learning*: This paper employs a learning mechanism known as Deep Q-learning, which is a hybrid of two reinforcement learning concepts that are extensively used: deep neural networks and Q-learning. Q-learning is a type of reinforcement learning that does not require a model [8]. It involves the assignment of a value, known as the Q-value, to an action that is executed from a specific state of the environment. The Q-value is denoted as [9]:

$$Q(s_t, a_t) = r_{t+1} + \gamma \max_A Q'(s_{t+1}, a_t) \quad (2)$$

where the value of the action a_t taken from state s_t is denoted by $Q(s_t, a_t)$. The term r_{t+1} within the parenthesis denotes the reward associated with making an action from state s_t . The Q-value of the immediate future is denoted by the term $Q(s_{t+1}, a_t)$, where s_{t+1} is the subsequent state in which the environment has evolved following the action taken in st. The term $\max_A$ denotes that the most advantageous action is chosen from the potential actions in state s_{t+1} . The discount factor, denoted by the symbol γ , is a value between 0 and 1, which diminishes

the significance of the future benefit compared to the immediate reward.

5) *Training Process*: In order to enhance the agent's performance and learning efficacy, experience replay [8] is implemented during the training phase. The process involves submitting the information required for learning to the agent in the form of a randomized group of samples, known as a batch, rather than promptly submitting the information that the agent gathers during the simulation, a process that is commonly referred to as "Online Learning." The batch is extracted from a data structure that is intuitively referred to as "memory." This structure is responsible for storing each sample that was collected during the training phase. A sample ρ is defined as:

$$\rho = s_t, a_t, r_{t+1}, s_{t+1} \quad (3)$$

The optimal traffic efficiency is accomplished by selecting the action with the highest value based on the current state after the deep neural network sufficiently approximates the Q-learning function. The decision is either to take exploratory action and potentially learn more or to take exploitative action and attempt to optimize the current knowledge about the environment's evolution.

III. PERFORMANCE ANALYSIS

In this section, the performance of the DDQN-based adaptive traffic signal cycle control is discussed. The SUMO-based intersection is a four-way intersection, with four lanes per arm approaching from the compass directions. The length of each limb is 750 meters. In particular, the left-most lane on each arm is equipped with a dedicated traffic signal, while the other three lanes share a traffic light. Every traffic light in the environment operates in accordance with common European regulations, with the sole exception being the absence of time between the conclusion of a yellow phase and the commencement of the subsequent green phase. In this setting, pedestrian crossings, sidewalks, and pedestrians are not accounted for. The training is divided into numerous episodes. The total number of episodes is 300. The total number of steps per episode is 5400, as SUMO defaults to a time-frequency of 1 second per step and a period of 1 hour and 30 minutes for each episode. The total training duration is approximately six hours on a high-end laptop, and 300 episodes of 1.30 hours each is equivalent to nearly 19 days of continuous traffic.

As illustrated in Fig. 3 the queue length drops to zero after the 4308th step. Since the queue length becomes zero, the waiting time for the vehicle is zero, and the difference in the CO₂ emissions from the current and previous steps is zero. This trend is further depicted in Fig. 4, which presents the reward function relative to the action steps. It can be observed that as the number of action steps increases, the reward function stabilizes to 0. This stabilization occurs because, once the model is

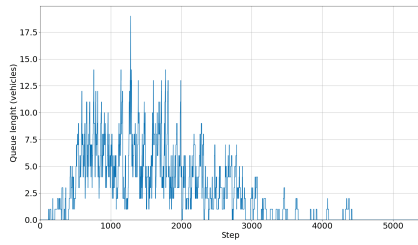


Fig. 3: Queue length with respect to testing step

fully trained, the network reaches a steady state, leading to minimal differences in both waiting time and CO₂ emissions.

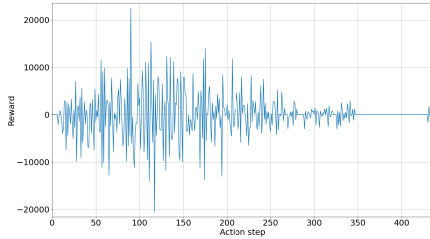


Fig. 4: Reward function with respect to action step

Fig. 5 shows the average waiting time with respect to the number of vehicles. It can be observed that as the number of vehicles increases, the average waiting time also increases. Specifically, the average waiting time of the vehicles increases by 15.58% as the number of vehicles increases from 1750 to 2000. A similar trend could be observed from Fig. 6 whereas the number of vehicles increases, the average CO₂ emissions also increase. Specifically, the average CO₂ emissions of the vehicles increase by 25% as the number of vehicles increases from 1750 to 2000.

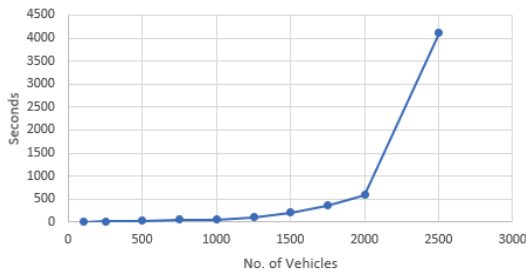


Fig. 5: Average waiting time with respect to the number of vehicles

IV. CONCLUSIONS

This paper proposes an adaptive traffic signal cycle control system for green city traffic management. We analyze the feasibility of DDQN to adapt the traffic signal cycle in response to varying network traffic conditions. A

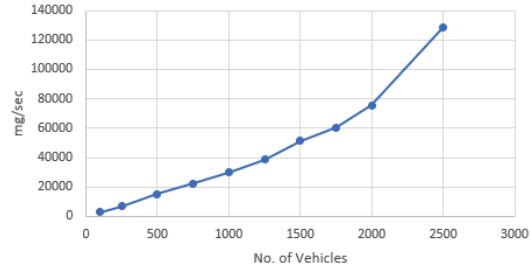


Fig. 6: Average CO₂ emission with respect to the number of vehicles

SUMO-based traffic simulation was employed to manage traffic signal cycles in smart cities. The proposed adaptive control system aims to reduce greenhouse gas emissions while also decreasing driver waiting times. Additionally, the evaluation of two reward metrics for agent actions has highlighted the importance of green city traffic management in mitigating the impact of traffic congestion on quality of life, fuel consumption, air quality, and human productivity.

V. ACKNOWLEDGEMENT

Dr. Vivek Bohara would like to thank TIH Foundation for IoT & IoE at IIT Bombay for the TiH-IoT CHANAKYA Faculty fellowship (TIH-IoT/2023-02/HRDCHANAKYA/SL/CFF-002) which partly supported his contribution in this work.

REFERENCES

- [1] K. Dubey and T. Gupta, "Adaptive traffic control system: The smart and imperative traffic control system for india," in *2020 International Conference on Intelligent Engineering and Management (ICIEM)*, 2020, pp. 91–96.
- [2] M. A. S. Kamal, T. Hayakawa, and J.-i. Imura, "Development and evaluation of an adaptive traffic signal control scheme under a mixed-automated traffic scenario," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 2, pp. 590–602, 2020.
- [3] Y. Chen and C. G. Cassandras, "Adaptive traffic light control for competing vehicle and pedestrian flows," in *2024 European Control Conference (ECC)*, 2024, pp. 1875–1880.
- [4] Y. Du, W. ShangGuan, D. Rong, and L. Chai, "Ra-tsc: Learning adaptive traffic signal control strategy via deep reinforcement learning," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, pp. 3275–3280.
- [5] T. Zhang, J. Xu, S. Cong, C. Qu, and W. Zhao, "A hybrid method of traffic congestion prediction and control," *IEEE Access*, vol. 11, pp. 36 471–36 491, 2023.
- [6] "Simulation of urban mobility." [Online]. Available: <https://eclipse.dev/sumo/>
- [7] A. Vidali, L. Crociani, G. Vizzari, and S. Bandini, "A deep reinforcement learning approach to adaptive traffic lights management," in *Workshop "From Objects to Agents" (WOA 2019)*, 2019. [Online]. Available: <https://ceur-ws.org/Vol-2404/paper07.pdf>
- [8] Z. Wang, T. Schaul, M. Hessel, H. Van Hasselt, M. Lanctot, and N. De Freitas, "Dueling network architectures for deep reinforcement learning," in *Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48*, ser. ICML'16. JMLR.org, 2016, p. 1995–2003.
- [9] "Deep Q-Learning, Part2: Double Deep Q Network, (Double DQN)," April 2019. [Online]. Available: <https://medium.com/@qempsiil0914/deep-q-learning-part2-double-deep-q-network-double-dqn-b8fc9212bbb2>